Analysis of Molecular MUX PUFs with Stochastic Challenges

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***Abstract*—Physical unclonable functions (PUFs) are small circuits primarily used as hardware security primitives for authentication. PUFs work by generating unique signatures due to the inherent random variations in the PUF manufacturing process. This paper addresses molecular PUFs that can be used for authentication in biosecurity applications where these PUFs can be naturally implemented using bio-substrate (as opposed to electronic). This paper analyzes a novel stochastic molecular multiplexer (MUX) PUF where the input challenge corresponds to a stochastic number encoded using a *fractional coding representation*; thus, the challenge is not binary. Each multiplexer in the PUF is realized using six reactions. The source of randomness in the PUF is the intrinsic variations in the *rate constants* of each reaction. Intra-PUF and inter-PUF variations are simulated in linear MUX PUFs consisting of 8, 16, 32, 64, and 96 stages. These variations are used to compute the reliability and uniqueness of different PUF configurations. It is shown that, for the stochastic molecular PUFs to be feasible, the distributions of the rate constants require large variances.**

***Index Terms*—Physical unclonable functions (PUF), molecular PUF, fractional coding, stochastic MUX PUF, stochastic molecular multiplexers, and biosecurity.**

I. Introduction

Physical unclonable functions (PUFs) are small circuits that are widely used as security primitives for hardware authentication. PUFs generate a unique signature from an input challenge due to unique process variations which are randomly created in the manufacturing of each individual PUF [1, 2, 3, 4]. PUFs are considered valuable security primitives due to their unique signatures and their inherent unclonability property [5, 6, 7]. Many types of PUFs have been presented previously; these include: delay PUF, radio-frequency PUF, and magnetic PUF [8, 9]. This paper examines multiplexer-based (MUX-based) PUFs, a subset of delay PUFs that are widely deployed.

Molecular computing is a computing paradigm that utilizes molecules and chemical reactions, instead of electrical signals and semiconductors, in order to perform computations [10]. Chemical reaction networks (CRNs) specify species of molecules and the relationships or reactions between these species. These CRNs can be used to easily design complex systems for computation [11, 12]. It has been shown previously that many basic computing building blocks can be implemented with CRNs such as AND gates, OR gates, and multiplexers [13]. CRNs have also been shown to be easily mapped to DNA strand displacement (DSD) reactions containing either one or two reactants [14]. The DSD reactions can be used to synthesize complex combinational and sequential logic circuits [3, 15, 16, 17].

Various types of PUFs have been presented previously in the literature, including the molecular MUX-based PUF first introduced in [18] where each MUX is realized using 16 molecular reactions. Furthermore, the challenges in this PUF are binary. In this paper, we introduce and analyze a novel molecular MUX-based PUF based on a set of previously published molecular reactions for a MUX first introduced in [13]. Each multiplexer, derived using the fractional coding scheme [19], requires 6 molecular reactions. This fractional coding scheme means that the multiplexer inputs and output are stochastic. This paper introduces intrinsic randomness into the PUF via changes in the rate constants of the molecular reactions, assuming this distribution to be Gaussian in nature. This is meant as a proof of concept to show the feasibility as a valid PUF, with further experimentation required to determine practicality. The PUF is constructed by cascading together a number of individually synthesized multiplexers according to the PUF configuration requirement. Five configurations, 8, 16, 32, 64, and 96-stages of PUFs are simulated and analyzed in this paper. The PUF presented in this paper is purely molecular and has not been mapped to DNA. The kinetics of CRNs can be obtained by simulating the systems of ordinary differential equations via the mass-action law to determine behavior.

The molecular MUX PUF with stochastic challenges is attractive as the number of reactions per MUX can be reduced from 16 to 6. However, we observe that these PUFs require larger variations in the rate constants for the PUFs to be feasible.

A single PUF must generate different responses under different challenges, and different PUFs must generate different responses under the same challenge. These ideals can be quantitatively measured using two PUF properties: $Reliability$ and $Uniqueness$. In addition, the distributions of inter-chip and intra-chip variations must not overlap in order for the PUF to be valid, i.e., the minimum inter-chip variation must be higher than the maximum intra-chip variation.

The remainder of this paper is organized as follows. Section II gives background on preliminary topics including two important PUF metrics: reliability and uniqueness. Section III describes the novel molecular PUF presented in this paper and the method by which random variations are introduced into the PUFs. In total, reliability and uniqueness are studied in PUF architectures of 8, 16, 32, 64, and 96 stages and these results are also presented in this section. Finally, Section IV concludes the paper.

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II. Background

A. DNA Fractional Coding

Fractional coding is an extension of stochastic computing [20, 21, 22] into the molecular computing domain [13]. Fractional coding works by first converting the desired logic function to the stochastic domain representation before converting it to a CRN. Each 'signal' is represented by two species: X_0 and X_1. The signal value $x \in (0, 1)$ in the unipolar format can be described as $x = \frac{[X_1]}{[X_0]+[X_1]}$.

CRNs with fractional coding have been used to synthesize support vector machines [23], artificial neural networks [24] and reservoir computing systems [25].

B. Multiplexer-Based Delay PUFs

MUX-based delay PUFs are a subset of delay PUFs that utilize two interconnected chains of multiplexers, each with randomness in propagation delay, to introduce variation into the PUF. A clock edge is input to the cascaded MUX chain. At each stage, the outputs either propagate straight or are swapped as inputs to the next stage depending on the control bit, referred as the challenge bit. The arbiter at the end of the chains produces either a 0 or 1 depending on whether the clock edge reaches the top or bottom input of the arbiter first.

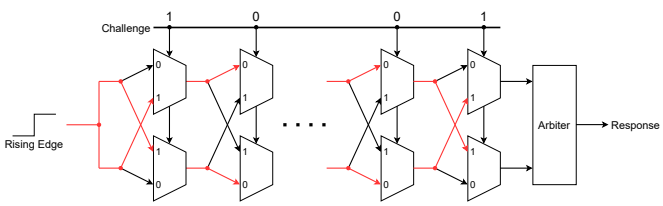


Fig. 1: Architecture of an N-stage MUX-based delay PUF.

1) Additive Linear Delay Model: These delay-based MUX PUFs can be represented using the linear additive delay model. For each stage in a MUX PUF, the delay can be modeled as an independent identically distributed (i.i.d.) random variable, D_i [2]. This variable follows a Gaussian distribution with mean μ and standard deviation σ, $\mathcal{N}(\mu, \sigma^2)$. The delay difference Δ_i at the ith stage can be described by ($\Delta_i = D_i^t - D_i^b$). The total delay difference after the final stage, R_N is described by ($r_N = \sum_{i=1}^{N}(-1)^{C'_i}\Delta_i$), with $C'_i = \oplus_{j=i+1}^{N} C_j$ and $C'_N = 0$. Finally, the response bit R is described as follows:

$$R = \begin{cases} 1, & r_N \geq 0, \\ 0, & r_N < 0. \end{cases}$$

The manufacturing process variations are what cause the variance in the MUX propagation delay which is what causes the arbiter output to vary, depending on the challenge. In the case of the molecular PUF, the rate constants are the sources of inherent randomness, which impact the propagation delay. This means that while all PUFs are produced the same way, they will still generate a different set of responses, or signatures, for a given set of challenges because of the randomness in the rate constants. Additionally, a single PUF instance may generate a different signature to the same set of challenges under significant enough noise conditions.

2) PUF Properties: We can quantify the properties described above using two PUF metrics: *Reliability and* Uniqueness [2]. These two metrics allow us to quantify the degree to which a single PUF produces similar responses to the same challenge under noise conditions and to which different PUFs produce similar responses to the same challenge, respectively. The PUF metric of reliability is computed using the intra-PUF variation as $Reliability = 1 - P_{intra}$, where $P_{intra} \in [0, 1]$ is the intra-PUF variation. The closer the value of $Reliability$ is to 1, the more reliable the PUF is. The PUF metric of uniqueness is computed using the inter-PUF variation as $Uniqueness = 1 - |2P_{inter} - 1|$, where $P_{inter} \in [0, 1]$ is the inter-PUF variation. The closer the value of $Uniqueness$ is to 1, the more unique the PUF is. More details about intra-PUF Variation, inter-PUF Variation, Reliability and Uniqueness can be found in [26].

III. Molecular MUX-based PUF with Fractional Coding

In this section, first the CRN built with fractional coding is outlined for a simple 2-to-1 multiplexer. This basic unit is then used to synthesize each of the various CRN MUX PUF architectures. For each of these architectures, the reliability and uniqueness are used to evaluate PUF performance.

A. CRN for a 2-to-1 MUX

The equations for the CRN MUX are listed in (1), with additional features such as reactant replication and output annihilation to ensure that the output of each MUX is stable and bounded. Before each authentication, the concentrations of the output species C_0 and C_1 are set to 0. By setting both the initial concentrations $[X_0]$ and $[X_1]$ to 100, the signal value of the 'rising edge' or input to the first stage is set to 0.5.

$$\begin{cases} A_0 + S_0 \xrightarrow{rate} C_0 + A_0 + S_0, A_1 + S_0 \xrightarrow{rate} C_1 + A_1 + S_0, \\ B_0 + S_1 \xrightarrow{rate} C_0 + B_0 + S_1, B_1 + S_1 \xrightarrow{rate} C_1 + B_1 + S_1, \end{cases} \tag{1a}$$

$$\{C_0 \xrightarrow{rate} \emptyset, C_1 \xrightarrow{rate} \emptyset, \tag{1b}$$

The primary feature that makes this MUX stochastic is the use of soft challenges. The challenge signals are represented using two different species, S_0 and S_1, whose concentrations together represent the soft challenge signal. This gives the input challenge a real-valued range on $(0, 1.0)$.

B. CRN for a PUF

Based on the configuration shown in Fig. 1, the molecular PUF is synthesized by constructing its component MUXes via the formal CRN described by (1). During this process, genuine or artificial manufacturing process variations impart minor variations in reaction rate constants for each of the reactions in each MUX. Similar to the delay differences within stages in silicon MUX PUFs, Δ_i, the rate constants follow a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$. In this paper, the Gaussian distribution

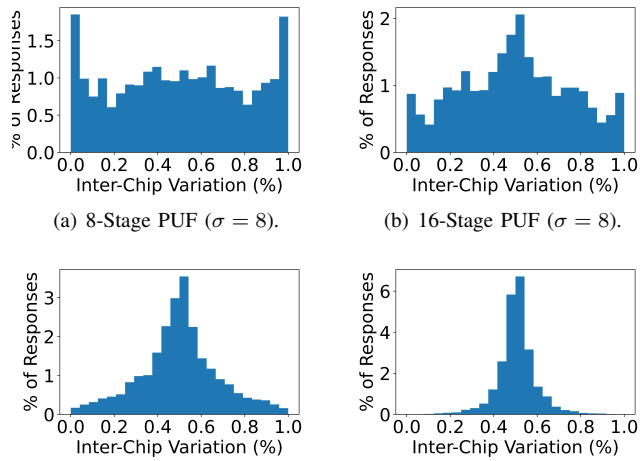
of $\mathcal{N}(16, 8)$ is sampled for all rate constants. This is higher than in previous CRN MUX-based PUFs such as in [18] but is necessary to obtain a working PUF. The absolute value of negative rate constants is taken to ensure all rate constants are positive. The response R only depends on the top and bottom output concentrations at the output of the final MUX stage, $C^t_{0,N}$, $C^t_{1,N}$, $C^b_{0,N}$, and $C^b_{1,N}$. Utilizing the fractional coding format, we can then compute the signal values of c^t_N and c^b_N. The response R is determined from the difference of the two final stage output signals, c^t_N and c^b_N, at equilibrium. If $c^t_N > c^b_N$, then R is 1; otherwise it is 0. This is the same operation as taking the sign of the expression $c^t_N - c^b_N$. Neither this operation nor the arbiter is synthesized in CRN. In this paper the final stage output concentrations, $C^t_{0,N}$, $C^t_{1,N}$, $C^b_{0,N}$, and $C^b_{1,N}$, are directly obtained from the simulated PUF, and the response is computed in software by taking sign of R defined as $R = \frac{C^t_{1,N}}{C^t_{0,N}+C^t_{1,N}} - \frac{C^b_{1,N}}{C^b_{0,N}+C^b_{1,N}}$.

1) Rate Constant Distribution: The rate constants used for the CRN equations within each MUX are sampled from a Gaussian distribution, $\mathcal{N}(\mu, \sigma^2)$. Previous work on CRN PUFs such as [18] used $\sigma^2 = 1$. A PUF output will have an amount of variation based on the inherently random stage delay differences and the number of stages that are present. The more stages there are to contribute to the variance, the more varied the output signal will be. We found $\sigma^2 = 1$ to not inject sufficient variance into each stage for the PUF to have a feasible uniqueness property. Standard deviations of $\sigma = \{2, 4, 8\}$ are tested to allow sufficient variance for the PUF to be unique. Plots of the inter-chip variation distributions for PUF architectures with $\sigma = 8$ are included in Fig. 2. We found that only $\sigma = 8$ provided sufficient variance in the rate constants to allow 64-stage PUFs to have viable uniqueness, as demonstrated in Fig. 2. Additionally, Fig, 3 demonstrates a single case of two 8-stage PUFs with two different sets of rate constants instantiated with $\sigma^2 = 8$, activated by the same set of challenges. These two PUFs have different responses, 0 and 1, showing that PUFs activated by the same challenge can produce different responses. The PUF Performance Metrics results presented in the following section are all gathered on PUFs whose rate constants are sampled from a $\mathcal{N}(16, 8^2)$ Gaussian distribution. The rate constants in CRNs are positive, and therefore the absolute value of all rate constants are taken to ensure that no negative rate constants are used for the PUF simulation results.

C. PUF Performance Metrics

The performance of the PUF architectures outlined in this section is evaluated using simulation software in order to systematically determine PUF viability. The results in section III-B showed the level of variance that is required for PUFs to be considered unique is substantial, and this subsection expands by calculating previously mentioned PUF performance metrics from the simulation results.

1) Reliability: This metric is measured by applying L different real-valued soft challenges $S \in \{0, 1.0\}$ to a single



(a) 8-Stage PUF ($\sigma = 8$). (b) 16-Stage PUF ($\sigma = 8$).

(c) 32-Stage PUF ($\sigma = 8$). (d) 64-Stage PUF ($\sigma = 8$).

Fig. 2: Inter-chip variation distributions for 8, 16, 32, and 64-stage PUFs using a rate constant standard deviation of $\sigma = 8$.

N-stage PUF m-times under varying noisy environments. This gives us m different L-bit long response vectors, R. The environmental noise is sampled from a Gaussian distribution $\mathcal{N}(0, \sigma_s^2)$. In this paper standard deviations of $\sigma_s = 0.05, 0.15, 0.3, 0.5$ are tested. Each of the $m = 500$ different response vectors is compared to all other response vectors, giving $\binom{m}{2} = \binom{500}{2} = 124750$ comparisons. A tabular format of the intra-chip variation results for PUFs of 8, 16, 32, 64, and 96-stages can be found in Table I, and the tabular results of the reliability for the same PUF architectures can be found in Table II.

2) Uniqueness: This metric is determined using K different PUFs, each with the same architecture and different rate constants which are sampled from the same Gaussian distribution, $\mathcal{N}(\mu, \sigma^2)$. In this paper, we used $K = 500$ different PUFs to generate 500 different responses, each of length $L = 200$ to evaluate the uniqueness metric for each of the PUF architectures. Each of the $K = 500$ different response vectors is compared to all other response vectors, giving $\binom{K}{2} = \binom{500}{2} = 124750$ comparisons. Tabular results for both the inter-chip variations and the uniqueness for PUFs of 8, 16, 32, and 64-stages can both be found in Table III.

Another pre-requisite for a viable PUF not yet mentioned is that the distributions of the inter-chip and intra-chip variations should not overlap. Overlapping means that it is probable that random noise could cause erroneous false verification in a real-world application. To prevent this, we must ensure that the maximum intra-chip variation does not surpass the minimum intra-chip variation [27]. Though, given that both of these quantities roughly model Gaussian distributions, it is generally sufficient to compare $\mu \pm 3\sigma$. In our specific case, this quantity calculated for the inter-chip variation is far below the maximum intra-chip variation for 8 and 16-stage PUFs, as the σ is so high the quantity becomes negative. This aligns

TABLE I: Intra-Chip Variation Results for PUFs of 8, 16, 32, 64, and 96 Stages

Noise	0.05			0.15			0.30			0.50		
Stages	**Max**	**Min**	**Mean**	**Max**	**Min**	**Mean**	**Max**	**Min**	**Mean**	**Max**	**Min**	**Mean**
8-Stage	0.00%	0.00%	0.00%	2.00%	0.00%	0.32%	7.00%	0.00%	2.91%	0.00%	0.00%	0.00%
16-Stage	3.00%	0.00%	0.78%	9.00%	0.00%	3.75%	19.50%	3.00%	10.50%	15.00%	2.00%	8.18%
32-Stage	2.50%	0.00%	0.60%	5.50%	0.00%	1.96%	11.00%	0.50%	5.06%	2.00%	0.00%	0.25%
64-Stage	3.50%	0.00%	1.08%	13.00%	1.00%	6.28%	32.00%	1.00%	20.89%	33.00%	10.00%	21.44%
96-Stage	7.50%	0.00%	3.10%	9.50%	0.00%	4.33%	24.50%	6.00%	14.62%	40.50%	16.00%	27.62%

TABLE II: Reliability Results for PUFs of 8, 16, 32, 64, and 96 Stages

	Noise			
Stages	**0.05**	**0.15**	**0.30**	**0.50**
8-Stage	100.00%	99.68%	97.09%	100.00%
16-Stage	99.22%	96.25%	89.50%	91.82%
32-Stage	99.40%	98.04%	94.94%	99.75%
64-Stage	98.91%	93.71%	79.11%	78.56%
96-Stage	96.90%	95.67%	85.38%	72.38%

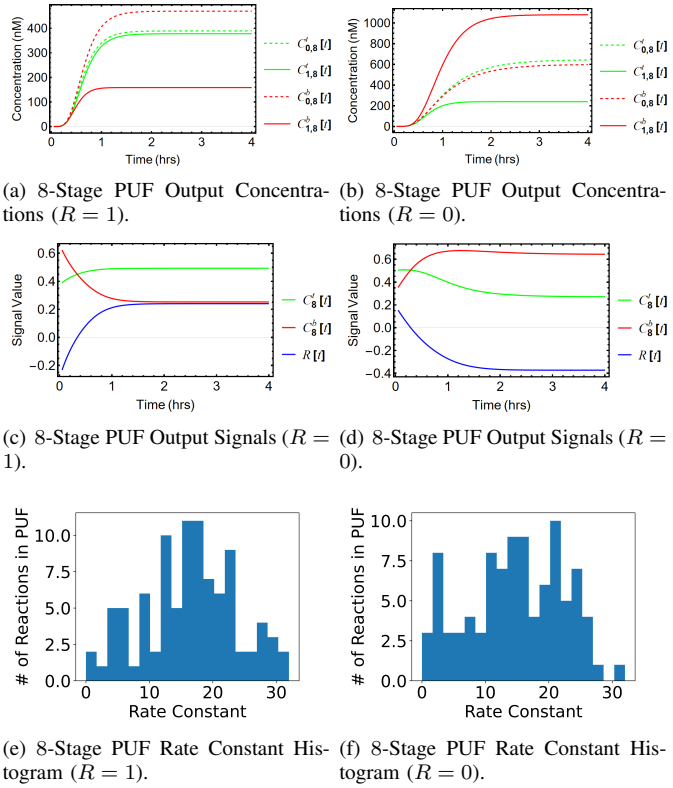
TABLE III: Inter-Chip Variation and Uniqueness Results for PUFs of 8, 16, 32, 64, and 96 Stages

	Inter-Chip Variation				
Stages	**Max**	**Min**	**Mean**	σ	**Uniqueness**
8-Stage	100.00%	0.00%	50.01%	30.9%	99.98%
16-Stage	100.00%	0.00%	50.00%	24.8%	99.99%
32-Stage	100.00%	0.00%	50.05%	15.9%	99.91%
64-Stage	99.50%	0.50%	51.42%	10.4%	97.15%
96-Stage	96.00%	2.50%	50.76%	6.9%	98.48%

with the distributions shown in Fig. 2, and the 8 and 16-stage inter-chip variation distributions for these architectures appear to be Gaussian with some uniformly distributed component, populating the entire range between 0 and 1, overlapping with the intra-chip distribution, meaning that these are not viable PUFs. Based on Fig. 2, the width of the distribution of inter-chip variations gets narrower from 16-stages through 64-stages, and as such we would expect the minimum inter-chip variation quantity to also increase, as is confirmed in Table III.

IV. Conclusion

We have investigated the viability of the stochastic molecular MUX PUF. As in previous work, [18], the rate constant variation has been assumed to both be possible and Gaussian in nature. The necessity of a molecular or DNA arbiter to complete the MUX PUF as well as the need for a mechanism to reset the concentrations of all intermediate molecular species to enable PUF re-usability limit the practicality of the PUF. The rate constants for each of the MUX's are sampled from a Gaussian distribution $\mathcal{N}(16, \sigma^2)$ in this paper, with σ values of 2, 4, and 8. The performance of the PUFs is based on these rate constant variances, as the more variance the PUF delay has the better the uniqueness will be. Through our analysis, we have determined that in order for this PUF to be viable, the standard deviation of rate constants should satisfy $\sigma \geq 8$. Previous DNA MUX PUF, such as that presented in [18] only required $\sigma \geq 1$. The benefit of this PUF over the result in [18] is the reduced number of reactions required for one MUX from 16 to 6. This requires a total of $12N$ reactions for an N-stage PUF, as each PUF requires $2N$ multiplexers; however, this requires significantly larger variances in the rate constants.



(a) 8-Stage PUF Output Concentrations ($R = 1$).

(b) 8-Stage PUF Output Concentrations ($R = 0$).

(c) 8-Stage PUF Output Signals ($R = 1$).

(d) 8-Stage PUF Output Signals ($R = 0$).

(e) 8-Stage PUF Rate Constant Histogram ($R = 1$).

(f) 8-Stage PUF Rate Constant Histogram ($R = 0$).

Fig. 3: Simulation results for two 8-stage PUFs with different rate constants activated by the same challenge. Challenge used is {0.478554, 0.00869692, 0.347029, 0.13928, 0.180603, 0.528701, 0.578587, 0.760349}.

We conclude that the stochastic molecular MUX PUF presented in this paper is only viable when the variance of the rate constant distribution is high. Whether these high variances can be achieved from a natural source of randomness remains unexplored. Future work needs to be directed towards exploring usefulness of the proposed structure as a feasible PUF using a natural source of randomness. New PUF structures that satisfy feasibility with reduced variance in rate constants also need to be investigated. There is also a need to demonstrate feasibility of DNA PUFs by mapping the molecular PUFs to DNA strand displacement reactions.

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